UNIT 5 · 8 hours

Lag–Lead Compensator



Compensators, Controllers and State-Space Analysis

Control Systems (22EEG53)
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Learning objectives

- Write the lag-lead compensator transfer function
- Explain why it combines the benefits of lag and lead
- Interpret its Bode plot
- Outline the design procedure

Syllabus context

Compensators and controllers: introduction to compensators, lag, lead and lag-lead compensators, P, PI and PID controllers.

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Why lag–lead?

- Lead alone: good transient but little help with steady-state error.
- Lag alone: good steady-state accuracy but a slower response.
- When both transient and steady-state specifications must be improved, use lag–lead.
- A single network provides phase lag at low frequency and phase lead at high frequency.

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Transfer function

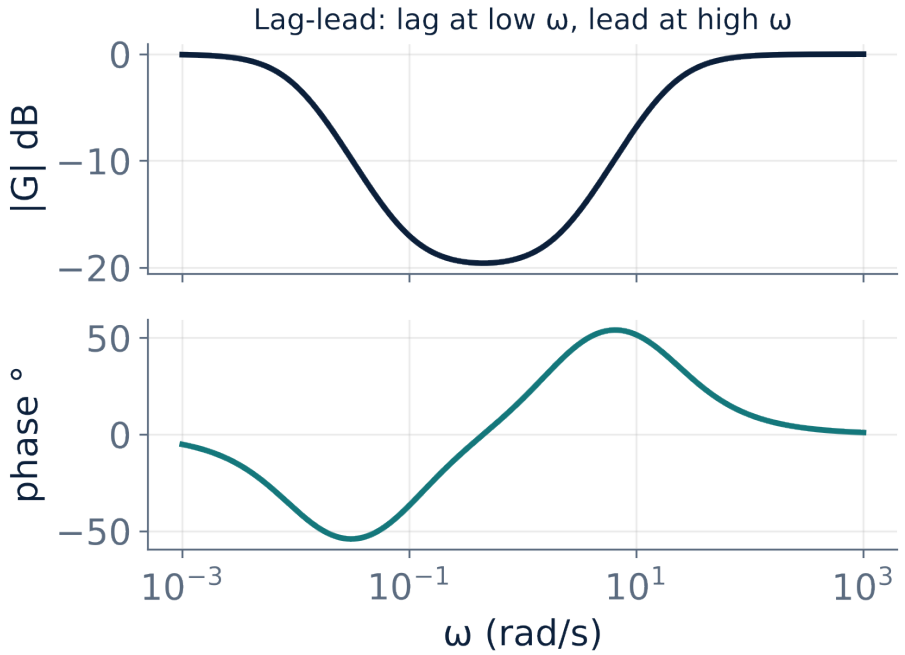
$$G_c(s) = [(1 + aT_1 s)/(1 + T_1 s)] \cdot [(1 + T_2 s)/(1 + \beta T_2 s)]$$

lead × lag, $a > 1$, $\beta > 1$

- Lead part: zero at $-1/(aT_1)$, pole at $-1/T_1$ (higher frequencies).
- Lag part: zero at $-1/T_2$, pole at $-1/(\beta T_2)$ (low frequencies).
- Often $a = \beta$ for a single passive network.

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Bode plot of a lag–lead network



- Low ω: negative phase (lag).
- High ω: positive phase (lead).
- Gain dips between the lag and lead corners.
- Example: lag β = 10, T2 = 10 s; lead a = 10, T1 = 0.05 s.

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Design outline

1

Choose K

Meet the steady-state error requirement.

2

Lead section

Design the lead part to obtain the required PM at the new crossover ωc.

3

Lag section

Place its corners at least a decade below ωc, so it does not disturb the phase margin.

4

Check

Verify PM, GM, bandwidth and steady-state error; iterate.

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Comparison

Lead / Lag / Lag-lead

- Lead: ↑ PM, ↑ bandwidth, noise up
- Lag: ↑ Kv, ↓ bandwidth, slower
- Lag-lead: both, more complex

Typical use

- Lead: fast servo, low-order plant
- Lag: process control, precise tracking
- Lag-lead: demanding systems with strict accuracy and speed

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Key takeaways

- Lag–lead = lag (low ω) × lead (high ω)
- Improves both steady-state and transient response
- Lead sets PM at the new crossover; lag sets Kv
- More components and design effort

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Practice questions

Q1. Write the transfer function of a lag–lead compensator.

Q2. Why are the lag corners placed a decade below the crossover?

Q3. When would you choose lag–lead over lead?

Q4. Sketch the Bode plot of a lag–lead compensator.